

MAHMOUD SALAMA

SOFTWARE ENGINEERING & APPLICATIONS LEADER

Engineering Leadership | Application Platforms | SDLC & Release Governance | Reliable Delivery
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PROFESSIONAL SUMMARY

Software engineering and applications leader with 18+ years across Egypt and Oman, progressing from hands-on Linux and .NET engineering to application-platform leadership, engineering governance and national-scale enterprise delivery. Leads software and application capabilities across a multidisciplinary technology organization, combining deep technical judgment with SDLC, QA/UAT, release, integration and production-readiness governance. Strong foundation across C#/.NET, web platforms, APIs, databases, reporting, integration and production operations.

CORE SKILLS

Software Engineering Leadership | Application Development & Modernization | Engineering Operating Model | SDLC & Release Governance | Architecture & Design Reviews | QA/UAT & Production Readiness | API & Integration Engineering | Application Security & Resilience | Technical Debt & Maintainability | DevOps / CI/CD Governance | Cross-Functional Engineering Leadership

SELECTED ENGINEERING IMPACT

- **Led the ~six-month end-to-end re-architecture and replacement of MediQ - winner of the Pharma Awards 2026 Digital Transformation Award - with a 54-person in-house technology organization.**
- Directed the completed **EGP 11M Microsoft Dynamics 365** integration across procurement, supply, inventory, finance and reporting, governing implementation quality, integration contracts, testing, security, change control and production transition.
- Established **API-led integration patterns** using REST APIs, middleware/integration services and governed database views, with authentication, mapping, reconciliation, master-data quality and performance controls.
- Built engineering delivery depth across **15+** Omani government platforms and **150+** major and mid-sized projects, progressing from hands-on full-stack development to leading development teams.
- Created **reusable** CMS, reporting and shared application patterns across **multilingual and RTL** solutions, supporting consistent delivery across multiple client and government projects.

PROFESSIONAL EXPERIENCE

Egyptian Authority for Unified Procurement (UPA)

05/2025 - Present

Head of the Central Administration for Information Systems & Digital Transformation

Cairo, Egypt

- **Lead engineering and application-platform direction across MediQ procurement, supply, inventory, tendering, reporting and connected enterprise services; architected cross-government integrations with ECIC, MoH/HMIS, GAHAC and Blood Bank systems while coordinating multidisciplinary engineering dependencies.**
- **Personally designed and developed production healthcare REST APIs integrating UPA with external health entities, implementing token-based authentication, structured JSON contracts, beneficiary/prescription/dispensing retrieval, GTIN-enabled product identification and standardized error handling.**
- **Lead architecture and design decisions** with relevant technical teams; use PoCs when needed, assess architecture risk, manage **technical debt** and balance maintainability, scalability, security, cost and delivery constraints.
- **Govern production architecture across ReactJS/Node.js services, C#/.NET APIs, PostgreSQL, Redis caching and Docker/containerized services; designed the centralized reusable reporting architecture and selected Telerik Reporting for parameterized reports and standardized PDF/Excel outputs.**
- **Lead technical delivery for MediQ E-Tender** and the completed **EGP 11M Dynamics 365** integration, ensuring implementation quality, testability, acceptance, production readiness and supportability.
- **Own engineering production-readiness controls** including capacity planning, **load/performance testing**, QA/release gates, monitoring/logging, incident escalation and **RCA**, backup/recovery and DR coordination.

Egyptian Authority for Unified Procurement (UPA)

10/2020 - 05/2025

General Manager - Systems, Applications & Technical Support

Cairo, Egypt

- Directed application modernization from established **.NET Framework/Telerik** solutions toward **ReactJS/Node.js** services, **C#/.NET** APIs, PostgreSQL, Redis, Docker and Apache Superset while preserving operational continuity.
- Strengthened **SDLC** execution through iterative planning, engineering standards, architecture/design and code reviews, **QA/UAT** and release gates, documentation, incident escalation, root-cause analysis and knowledge transfer.
- Expanded engineering capability as workload grew, strengthening **QA/testing**, **DevOps**, **mobile** and application functions while connecting requirements, architecture, engineering, deployment, training and production support.
- Converted manual, paper- and spreadsheet-heavy processes into **governed digital workflows** with role-based access, approvals, validation, audit trails, reporting and defined operational ownership.

Integral Solutions LLC

05/2016 - 10/2020

Development Team Leader & Senior Full-Stack Developer

Muscat, Oman

- Led **10+ developers** while remaining hands-on across **15+** Omani government and e-government platforms, covering requirements, solution design, development, testing, deployment, releases and production support.
- Contributed extensively to the National Educational Portal serving **approximately 1 million users**, Maktabi Electronic Correspondence Management and the **Sultan Qaboos Award-winning** Ministry of Education Recruitment System.
- Designed and delivered secure government workflows, reporting, database and integration components across **C#/.NET**, ASP.NET Web Forms, **SQL Server** and **MySQL**.

Keyframe Egypt

04/2011 - 05/2016

Development Team Leader & Senior Full-Stack Developer

Cairo, Egypt

- Delivered or led **150+** major and mid-sized projects across government, healthcare, education, construction, automotive, events and commercial sectors, excluding small projects; progressed from hands-on engineering to leading a **six-person** development team.
- Owned analysis, **C#/.NET** engineering, database/reporting design, testing, deployment and support across **multilingual and RTL** solutions while balancing concurrent client priorities.
- Delivered **Yashfeen**, an **ITIDA-recognized** healthcare solution, and created reusable CMS, reporting and shared application patterns across multiple projects.

Egyptian Armed Forces

04/2008 - 10/2010

Linux Systems & Web Developer

Egypt

- Built hands-on Linux systems expertise through **Linux From Scratch**, kernel modules, optimization and hardening, alongside secure **ASP.NET/SQL Server** application development.

ENGINEERING LEADERSHIP & TECHNICAL TOOLKIT

Engineering Leadership: Software engineering management, application development, multidisciplinary team leadership, estimation and capacity planning, architecture/design reviews, SDLC governance, QA/UAT, release management, production support, technical debt and continuous improvement.

Modern Engineering: C#/.NET, ASP.NET, ReactJS, Node.js, REST APIs, PostgreSQL, SQL Server, MySQL, Redis, Docker, Git/TFS, Jenkins, CI/CD governance, Postman, Swagger/OpenAPI.

Full-Stack Foundation: ASP.NET Web Forms, .NET Framework, Entity Framework, LINQ, JavaScript, jQuery, AJAX, HTML/CSS, Bootstrap, REST/SOAP, Telerik, Kendo, Infragistics, Crystal Reports, Umbraco and Kentico.

SELECTED ENGINEERING & GOVERNANCE CONTRIBUTIONS

- National Pharmaceutical Track & Trace Program (2021-2024) - contributed to multi-agency governance, compliance and interoperability.
- Board Member - National Blood Operations Oversight (2021-2024), representing UPA and contributing to digital governance, data integrity and technology standards.
- Member - Egypt Digital Transformation Committee supporting the World Bank Group B-READY Assessment (2025-2026).

EDUCATION & LANGUAGES

B.Sc. in Computer Science, Faculty of Computers & Informatics, Zagazig University | 2003-2007

Arabic - Native | English - Professional Working Proficiency